Prodigy Anesthesia Quick Review

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What is the most common critical incident in anesthesia?	The most common critical incident in anesthesia is due to breathing circuit disconnections which most commonly occur at the y-piece. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 43.*
How is the leak test performed on an anesthesia machine?	The leak test is performed by closing the pop-off valve, occluding the y-piece, and evaluating for a drop in pressure after pressurizing the circuit to 30 cm H2O using the flush valve. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 215-216.*
On an anesthesia machine, what components are found in the low pressure system?	The low pressure system on an anesthesia machine contains the flowmeters, hypoxia prevention devices, unidirectional valves, pressure relief devices, and the common gas outlet. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 67-77.*
What are the two components of the circle system test on an anesthesia machine? Between what two machine components does the circle system test?	The circle system test evaluates the circle breathing system from the common gas outlet to the y-piece and consists of two parts, the leak test and the flow test. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 215-216.*
How is the flow test performed on an anesthesia machine?	The flow test checks the integrity of the unidirectional valves by removing the y-piece and breathing through each individual hose to observe the movement of each valve. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 215-216.*
On an anesthesia machine, what components are found in the intermediate pressure system?	The intermediate pressure system of an anesthesia machine contains the pneumatic part of the master switch, pipeline inlet connections and pressure indicators, the gas power outlet, oxygen flush valve, oxygen pressure failure devices, and flow control valves. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 59-66.*
Where does the oxygen flush receive its gas supply?	The oxygen flush receives an unmetered oxygen supply from the cylinder pressure regulator or the pipeline inlet at a flow rate between 35 and 75 liters/minute. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 62.*

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When is use of the oxygen flush control valve contraindicated in a mechanically ventilated patient?

The oxygen flush control valve delivers oxygen at a rate of 35-75 L/min and a pressure of 50 psig. To avoid barotrauma, it should never be used when there is no outlet for excess gas pressure to escape. This situation exists when the patient is in the inspiratory phase of mechanical ventilation when the inspiratory valve is open to the patient but there is no vent for gases to escape.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 621.

What are the three functions of the hanger-yoke assembly on the anesthesia machine?

The hanger yoke assembly orients the gas cylinder, supports the weight of the cylinder, provides for uni-directional gas flow, and provides an air-tight seal.

Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 53.

How does a variable-bypass vaporizer work?

A variable-bypass vaporizer splits the fresh gas flow into two parts. One portion enters the vaporizing chamber where it becomes saturated with anesthetic agent. The other portion enters the bypass chamber and does not come into contact with the anesthetic agent. By controlling the amount of gas that enters the vaporizing chamber, the amount of anesthetic agent delivered to the patient can be controlled.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia, PA: Elsevier Saunders; 2011: 202.

Is the Tex-6 vaporizer a variable-bypass vaporizer? Why or why not?

The Tec 6 vaporizer is heated to 39 degrees Celsius and pressurized to 2 atmospheres. It is not a variable bypass vaporizer and unlike these devices, no fresh gas flows through the sump.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 63.

What is the function of the bimetallic strip in a contemporary vaporizer?

The bimetallic strip is temperature sensitive and expands and contracts in response to ambient temperature changes to alter the total gas flow between the vaporizing and bypass chambers. If the temperature of the liquid anesthetic decreases, the bimetallic strip allow more gas flow into the vaporizing chamber to compensate for the decreased vapor pressure of the cooler liquid.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 203.

Which vaporizer is electrically heated?

The desflurane vaporizer is electrically heated. The vaporizers for other contemporary gases are temperature compensated and have a variable-bypass, flow-over design.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 202.

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What is the 'pumping effect' and when might it occur?	The 'pumping effect', an intermittent back pressure caused by positive pressure ventilation or use of the oxygen flush valve results in elevated vaporizer output. It is more prominent with low flows, low vaporizer dial settings, low levels of anesthetic in the vaporizer chamber, high respiratory rates, and high peak inspiratory pressures. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 664-665.*
What color is the CO2 absorbent pH indicator ethyl violet when it is fresh? When it is exhausted? Phenolpthlalein? Clayton yellow? Ethyl Orange? Mimosa Z?	Ethyl violet Fresh: White Exhausted: Purple Phenolpthalein Fresh: White Exhausted: Pink Clayton Yellow Fresh: Red Exhausted: Yellow Ethyl Orange Fresh: Orange Exhausted: Yellow Mimosa Z Fresh: Red Exhausted: White *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 139.*
Why should you avoid using a dessicated CO2 absorbent with desflurane?	Prolonged exposure of desflurane or isoflurane to dessicated CO2 absorbent can result in the degradation of the anesthetic with the resulting production of carbon monoxide. Increased temperature and higher doses of anesthetic agent both increase the amount of carbon monoxide produced. The inhaled anesthetic that produces the greatest amount of carbon monoxide is desflurane. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 641.*
When an anesthesia machine is connected to a pipeline gas supply, should the backup cylinder valves be open or closed? Why?	When connected to a pipeline gas source, the cylinders should be kept closed so that a pipeline pressure failure would be detected earlier and to prevent pressure variations in the pipeline from draining oxygen from the cylinders. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 60.*
How many liters of oxygen will an E-cylinder hold? Nitrous oxide? Carbon dioxide? Air?	An E-cylinder will hold about 660 liters of oxygen, 1590 liters or either nitrous oxide or carbon dioxide, or 625 liters of air. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 8.*
Which gas flowmeter should be positioned last in the anesthesia manifold and why?	The oxygen flowmeter should be positioned last in the sequence of flowmeters in the manifold to reduce the likelihood that a leak in one of the flowmeters would allow oxygen to escape and lower the inspired oxygen concentration. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 201.*

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Pressing the oxygen flush valve allows oxygen to bypass what components?	Pressing the oxygen flush valve creates a direct communication between the low and high pressure circuits and allows oxygen to bypass the flowmeters and manifold on its way to the outlet. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 202.*
A leak in the anesthesia machine due to the vaporizer is most commonly caused by what?	A loose filler cap. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 203.*
Running a high flow of nitrous oxide can cause frost to develop on the cylinder and valves. Why does this occur?	The vaporization of a compressed or liquified gas absorbs heat. This heat is taken from the components of the cylinder and the surrounding air resulting in the accumulation of frost. Frost does not form on the inside of the cylinder because it contains no water vapor. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 201.*
What is the heat of vaporization?	The heat of vaporization of a liquid is the number of calories required at a certain temperature to convert one gram of the liquid into a vapor. (It is the amount of energy required for a liquid to convert into a vapor.) As a liquid such as a volatile anesthetic or nitrous oxide converts into a gas, it loses heat and cools. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 202.*
What safety mechanisms are built into the pipeline inlet connections on the anesthesia machine?	The pipeline inlet connections on an anesthesia machine are fitted with diameter index safety system fittings to prevent connection of a gas line to the wrong connector on the machine. They also have 100 micron (or less) filters and a unidirectional check valve. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 24.*
What is the definition of boiling point?	The temperature at which the vapor pressure of a liquid is equal to the atmospheric pressure. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 80-81.*

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How does the oxygen supply failure alarm work?

The oxygen supply failure alarm sets off an alarm if the oxygen pressure falls below a standard set by the manufacturer (which is typically 30 psig). It must engage within 5 seconds of the disconnect and cannot be disabled by the anesthetist. It is designed to help prevent hypoxia from lack of oxygen flow, but does not prevent the flow of anesthesia gases (and therefore could still possibly allow a hypoxic mixture of gases to be delivered).

Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 61-62.

In the event your oxygen supply fails and you must switch indefinitely to your backup oxygen tanks, what are appropriate measures to reduce the amount of oxygen consumed and prolong the duration of your backup oxygen supply?

Most anesthesia machines utilize oxygen as the driving gas to power the ventilator. By reducing fresh oxygen flow rates and eliminating the use of the ventilator by allowing the patient to breathe spontaneously or ventilating via the reservoir bag, you will prolong the backup oxygen supply.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 621.

What is the function of the diameter index safety system?

The diameter index safety system connects the wall outlet gas supply to the anesthesia machine and has a line pressure of 45-55 psi. It is designed for medical gas lines at pressures of 1380 kPa (200 psi) or less

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 49-51.

What is the function of the nitrous oxide shutoff valve on the anesthesia machine?

The nitrous oxide shutoff device (also known as the fail safe device) triggers an alarm and closes the supply of nitrous oxide and other gases when the oxygen pressure falls below 30 psi.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 199.

Which gas should be positioned last in the flowmeter sequence on an anesthesia machine?

Oxygen should be placed last in the sequence of flowmeters to reduce the risk of delivery of a hypoxic mixture of gases if a leak develops in one of the upstream flowmeters.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia, PA: Elsevier Saunders; 2011: 201.

Why is the flowmeter tube (also called a Thorpe tube) tapered?

The diameter of the flowtube is tapered such that it is smaller at the bottom than at the top. At lower gas flows, it takes less pressure to support the bobbin and the diameter is smaller. At higher gas flows, the diameter is larger and more gas is allowed to pass around the bobbin.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 54-55.

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How should a flowmeter indicater be assessed?

If using a bobbin, the flow rate should be read at the top of the indicator. When using a ball-shaped float, the flow rate should be read at the middle of the ball.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 201.

How do you calculate the inspired oxygen concentration if you are administering air and oxygen simultaneously?

The formula to calculate the inspired oxygen concentration if you are administering both air and oxygen together is: 100 X (1.0 X O2 L/min) + (0.21 X Air L/min)/(O2 L/min + Air L/min)

Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 25.

You are testing your anesthesia machine in the morning before cases begin. You disconnect the pipeline supply and turn on the backup oxygen cylinder to make sure it functions correctly. The oxygen cylinder is full and connectly correctly but the anesthesia machine still does not pressurize. What could be the cause?

If you have verified that the backup oxygen cylinder is full, connected properly, and opened and the anesthesia machine still does not pressurize, then there is a problem in the high or intermediate pressure system in the anesthesia machine.

Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: 407.

What is the only machine safety device that detect problems downstream from the flowmeters?

The oxygen monitor is the only machine safety device downstream from the flowmeters. The fail-safe valve, proportioning system, and the O2 supply alarm are upstream from the flowmeters.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia, PA: Elsevier Saunders; 2011: 216.

What is the result of closing the APL valve off when the patient is breathing spontaneously?

Closing the APL valve while a patient is breathing spontaneously will result in an increase in the breathing circuit pressure. The breathing bag can become distended and the patient can be exposed to CPAP.

Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 118.

A patient is being mechanically ventilated with a respiratory rate of 8/minute, a tidal volume of 700 mL, the oxygen flow rate is 1L/minute and nitrous oxide is being delivered at 1L/minute. How much oxygen is being consumed per minute?

The ventilator is driven by oxygen and requires an amount equal to the minute ventilation, which in this case is 5.6 L/min. The oxygen flow is 1 L/min and therefore, the total flow consumed is 6.6 L/min.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 76-77.

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What characteristics of synchronized intermittent mandatory ventilation are mandatory?

SIMV delivers a set respiratory frequency that are synchronized with the patient's spontaneous inspiratory activity, if present. The breaths may be delivered at irregular intervals. A mandatory tidal volume and ventilatory rate must be selected. It can also be combined with pressure support ventilation.

Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: 318.

Which mechanical ventilation modes allows for spontaneous ventilation?

Ventilation modes that allow spontaneous ventilation include: IMV, SIMV, MMV, PSV, and HFJV. CMV, AC, and PCV do not.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1291.

When the bag/vent selector is set to Vent Mode, what components are eliminated from the breathing circuit?

The reservoir bag and the APL valve.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 209.

What change would you see if there was a leak in the bellows in an oxygen-driven ventilator?

Oxygen driven ventilators use a 50 psi pressure source to drive the ventilator. If a leak develops in the bellows then the patient is exposed to that higher pressure and peak inspiratory pressure would rise. The fraction of inspired oxygen would also increase.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 209.

What is the first step you should take if you notice hypoventilation in a mechanically ventilated patient?

The first step is to switch to manual ventilation. If there is an obstruction, you should switch to a resuscitation bag. If there is still a persistent obstruction then the fault must lie in the airway device or in the respiratory tract.

Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 224.

What are the regulations concerning the amount of pressure a breathing bag can exert? Who sets these regulations?

According to the standards set by the ASTM, 3-liter reservoir bags (also called breathing bags) that are distended to four times their normal size should not exert a pressure less than 35 cm H2O or greater than 60 cm H2O. Bags that are 1.5 L in size or smaller shall not exert a pressure less than 30 cm H2O or greater than 50 cm H2O when distended to this extent. The lowest pressure to be exerted is to ensure that the bag is capable of exerting enough minimum pressure to ventilate a patient and the highest pressure is to limit the amount of barotrauma that a patient may be exposed to when using the breathing bag.

Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: 196.

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What is wasted ventilation? Name four factors that will increase the amount of tidal volume lost during inspiration due to wasted ventilation.

Wasted ventilation is a reduction in the tidal volume delivered to the patient due to distention of the breathing system components. The tidal volume lost increases with increased airway pressures, increased tidal volume, increased compliance of the breathing system components, and increased breathing system volume.

Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 113.

What factors result in higher tidal volumes than are set for the patient when the anesthesia machine ventilator is in use?

Unless you are using an anesthesia machine that is designed to compensate for it, when the ventilator is being used a portion of the fresh gas flow will contribute to the tidal volume. This effect increases if the fresh gas flow rates are high, if the I:E ratio is high, and if the patient has a slower respiratory rate.

Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 171.

What are the potential complications of overfilling a vaporizer?

Possible complications from overfilling a vaporizer include: liquid agent entering the fresh gas line and potentially delivering a lethal dose of agent to the patient or complete failure of the vaporizer. Some vaporizers have the filling port positioned so that overfilling cannot occur, but if the vaporizer is tipped overfilling is still a possibility.

Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 103.

Describe the steps of the chemical reaction that occur in the soda lime-based CO2 absorber cannister.

The reaction that occurs within the CO2 absorber between soda lime and carbon dioxide occurs in two steps. The first is CO2 + H2O --> H2CO3. The second portion of the reaction is H2CO3 + 2NaOH --> Na2CO3 + 2H2O + heat. Therefore, H2CO3 is an intermediate product of the reaction, but not a final product as it is degraded by the secondary reaction. The final products are Na2CO3, H2O, and heat.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 37.

What is channeling in a CO2 absorber cannister?

Channeling occurs when gases flow through passages of lower resistance in the CO2 cannister. When this occurs, most of the CO2 absorbent granules are bypassed and the efficiency of the absorber decreases. It can be minimized by gently shaking the cannister prior to use.

Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia, PA: Elsevier Saunders; 2011: 213.

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How is resistance to air flow related to the size of the granules in a CO2 absorber?	As the particle size decreases, the resistance to gas flow increases. The particle size most commonly used is 4 to 8 mesh, which optimizes gas flow resistance with surface area for CO2 absorption. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 675.*
What are the advantages/disadvantages of a single-cannister CO2 absorber system over a double-cannister system?	A single canister CO2 absorber system is easier to change, but double canister systems provide more complete CO2 absorption, less frequent canister changes, and lower resistance to gas flow. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 38-39.*
What is the maximum amount of CO2 that soda lime can absorb? Calcium hydroxide lime?	Soda lime is capable of absorbing 26 liters of carbon dioxide per 100 grams of absorbent. Calcium hydroxide lime can absorb 10.2 liters of carbon dioxide per 100 grams of absorbent. The absorption efficiency is often much less due to channeling. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 675.*
Which preserves circuit humidity more effectively, a single-limbed circuit or a two-limbed circuit?	A single-limbed circuit will improve humidification to a slightly greater degree than a two-limbed circuit. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 212.*
In a circle anesthesia circuit, what determines the amount of gas that is rebreathed by the patient?	In a circle anesthesia circuit, the fresh gas flow rate determines the amount of gas that returns to the patient and is rebreathed. *Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 26.*
How much carbon dioxide will a patient inhale using a normal circle anesthesia circuit?	None. A circle circuit incorporates the carbon dioxide absorbers which will absorb any exhaled CO2. *Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011:*
You are using a circle anesthetic circuit and realize that the carbon dioxide absorbent has become exhausted. You decide to convert the circuit to an 'open circuit'. What is the easiest way to do this during the case?	An 'open' circuit implies that the patient is not rebreathing any of the exhaled gases. Although this is wasteful, it can also be useful in the event the CO2 absorbent becomes exhausted during a case and you need to prevent the patient from rebreathing exhaled carbon dioxide. All that's necessary to convert a circle circuit to 'open' is to increase the fresh gas flow rate to above the patient's minute ventilation. *Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 26.*

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Can you use both an HME and a heated humidifier at the same time to preserve circuit humidity?	No. The combination of an HME and a heated humidifier can produce a dangerous resistance to airflow. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 212.*
How will a PEEP valve affect the patient's work of breathing?	A PEEP valve will increase the amount of work the patient has to exert during the exhalation phase. *Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: 200-201.*
What is the Bain circuit?	The Bain circuit is a modification of the Mapleson D circuit utilizing a coaxial system in which the fresh gas hose lies within the corrugated tubing through which exhaled gas flows. The exhaled gas passes in the opposite direction of the fresh gas, establishing a countercurrent, which helps warm the inspired gas. It requires a fresh gas flow of 1-2 times that of the minute ventilation to prevent rebreathing. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 36.*
How can you prolong the life of an electrochemical oxygen sensor?	The key to prolonging its lifespan is reducing the amount of oxygen it is exposed to. It should be removed from the circuit and exposed to room air when not in use. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 424.*
Compared to the standard Bain circuit, what parameters are altered in the longer modification of the Bain circuit designed for use in an MRI suite?	Compared to a standard Bain circuit, the longer Bain modification exhibits increased static compliance, PEEP, and resistance to spontaneous respiration. If the ventilator settings are kept the same, the peak inspiratory pressure and tidal volume are decreased. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 189.*
The Bain circuit is a modification of what other anesthetic breathing system?	The Bain circuit is a coaxial version of the Mapleson D system. The fresh gas supply tube passes inside the corrugated expiratory tubing. The Bain circuit may be used for controlled or spontaneously breathing patients. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia, PA: Elsevier Saunders; 2011: 206.*
Which anesthetic breathing circuit does not have a gas reservoir bag?	The Mapleson E does not have a gas reservoir bag. The Bain circuit and all of the Mapleson circuits except for the Mapleson E have a gas reservoir bag. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia, PA: Elsevier Saunders; 2011: 205.*

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How can pain affect oxygen saturation during low-flow anesthesia?

During low-flow anesthesia, the amount of oxygen delivered must be higher than the patient's metabolic demand regardless of the FiO2. For example, if your oxygen flow rate is 300 mL/min and the patient's metabolic demand is 300 mL/min, then the oxygen supply meets the demand. If however, the patient experiences pain, it is likely that his heart rate, blood pressure, and metabolic demand for oxygen will increase. As his oxygen consumption increases above 300 mL/min the supply no longer meets the demand and the oxygen saturation will begin to fall.

Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 29.

What is a heat and moisture exchanger (HME)? What are three disadvantages of using an HME?

Heat and moisture exchangers (HME) are inexpensive, silent, easy-to-use devices that help retain heat and moisture within the anesthesia circuit. Advantages of HMEs include: no need for water or electrical power source and no risk of: hyperthermia, overhydration, burns, or electrical shock. They all act as large particle filters and many serve as effective bacterial and viral filters.

The primary disadvantages of HMEs are that they are not nearly as effective at warming and humidifying the patient's airway as water-based, electrical devices. They also increase deadspace and can increase the work of breathing.

Airway obstruction can occur if the HME becomes blocked with fluid, blood, secretions, nebulized drugs, or if a mechanical defect is present in the device. If increased airway resistance is experienced during an anesthetic, the peak pressure should be measured both with and without the HME in place.

Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 208-213.

What driving pressure is required to deliver adequate oxygen flow through a 14 or 16 gauge catheter used for an emergency cricothyrotomy?

When performing transtracheal jet ventilation through a 12 or 14 gauge catheter, a driving pressure of 50 psi is required to ensure sufficient gas flow into the lungs.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1237.

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What are the two types of heat and moisture exchangers (HMEs) and how are they different?	There are two primary types of HMEs: hydrophobic and hygroscopic. Hydrophobic HMEs have a hydrophobic membrane with small pores that is pleated to provide a greater surface area. Hygroscopic HMEs contain a wool, foam, or paperlike material that is coated with a chemical that helps it to retain moisture. High ambient temperature may decrease the effectiveness of hydrophobic HMEs compared to hygroscopic HMEs. Hydrophobic HMEs may also be more effective at preventing the transmission of the hepatitis C virus than hygroscopic HMEs. Also, if hygroscopic HMEs become wet, they may lose their ability to filter airborne pathogens and airway resistance may increase substantially. The airway resistance of hydrophobic HMEs increases only slightly if wet. Hygroscopic HMEs are considered more effective at preserving heat and humidity than hydrophobic HMEs. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 209-211.*
In what cases should an HME be avoided?	An HME should be avoided when a patient exhibits thick, copious secretions which could enter the circuit and occlude the HME. They should also not be used when there is a leak that prevents the exhalation of gas through the HME such as a bronchopleuralcutaneous fistula or when an uncuffed endotracheal tube is used. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 211.*
Can you use two HME's in a series to improve humidification in an anesthesia circuit?	Yes. Using two HME's in a series can improve humidification, but care must be taken to monitor for the effects of increased deadspace or work of breathing. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 211.*
How does the color of a gas cylinder correspond with its contents in the US?	Oxygen cylinders are green, nitrous oxide cylinders are blue, carbon dioxide cylinders are gray, air cylinders are yellow, nitrogen cylinders are black, and helium cylinders are brown. These colors can vary outside of the United States. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 9.*
An E-cylinder oxygen tank pressure gauge reads 950 psig. How much oxygen is left in the tank?	A full E-cylinder oxygen tank holds 660 liters at a pressure of 1900 psig. Because an oxygen tank's pressure is proportional to its contents, if the pressure falls by half, then the number of liters falls by half as well. Using this ratio, the tank would have 330 liters left in it. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 248-249.*

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Anesthetic Delivery Systems

How does an oxygen concentrator work and how much oxygen can it deliver?	An oxygen concentrator utilizes a technology called pressure swing adsorption which filters out nitrogen, carbon dioxide, carbon monoxide, water, and hydrocarbons while allowing oxygen and trace gases such as argon to pass through. The FiO2 delivered ranges between 90 and 96 percent. Although argon concentrations can reach as high as 5%, there are no known detrimental effects from short-term or long-term exposure to argon. High humidity can reduce the oxygen concentration delivered by a concentrator. A concentrator requires a compressor and the device can stop delivering oxygen if the compressor fails. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 20.*
What is the critical temperature of nitrous oxide in degrees Celsius? What is the significance of this?	The critical temperature of nitrous oxide is 36.5 degrees Celsius. This means that because its critical temperature is above room temperature, it will remain in liquid form without a refridgeration system. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 12.*
When the pressure gauge drops below 745 psig on a nitrous oxide E-cylinder at room temperature, how much nitrous oxide is left in the tank?	Because E-cylinder nitrous oxide tanks contain nitrous oxide in both the liquid and gas state, the only accurate way to determine the amount of gas left in the tank is by weighing it. A full E-cylinder tank will hold 1590 liters of nitrous oxide. When the liquid form is consumed and the tank pressure drops below 745 psig, however, the amount of nitrous oxide in the gas phase is about 400 liters. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 12-13.*
What is Entonox?	Entonox is a 50:50 mixture of oxygen and nitrous oxide. It is most commonly used for dental procedures, dressing changes, and obstetric anesthesia. *Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 13-14.*
What is heliox?	Heliox is a mixture of helium and oxygen. Helium is less dense than oxygen and improves respiratory mechanics by increasing laminar flow (a reduction in Reynolds number) through the airways and decreasing the work of breathing. Heliox can alleviate the symptoms of increased airway resistance, but doesn't dilate the airways like a bronchodilator. *Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 14.*

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